# Hiten Dalmia

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## Education

**Chennai Mathematical Institute** — Chennai, India
*Master of Science in Theoretical Computer Science (On Full Scholarship)* — *Aug. 2026 – Discontinued*

**Kirori Mal College, University of Delhi** — Delhi, India
*Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics* — *Nov. 2022 – May 2026*

## Experience

**Data Engineer** — Mar. 2023 – Present
*Freelance & Fractal* — *Remote*

- Delivered 10+ production web and data applications for client businesses via Upwork and Fractal, specializing in data engineering pipelines, backend APIs, and automated testing.
- **Order Procurement & Management Solution** | *Python, SQL, PostgreSQL, NextJS*
  - Deployed an enterprise order portal using NextJS, TypeScript, and PostgreSQL with RESTful APIs, session auth, and audit logging for 120+ monthly orders.
  - Designed a dual-tier analytics interface with a customer ordering portal and an administrative dashboard for real-time inventory tracking, staff dispatch, and recurring orders.
  - Developed a batch ETL/ELT data ingestion pipeline parsing incoming message streams into structured PostgreSQL tables with validation checks, eliminating manual entry.
  - Designed relational PostgreSQL database schemas with B-tree indexing and integrity constraints to handle 50,000+ records.

**Computer Vision Engineer** — Oct. 2023 – Apr. 2025
*Tagglabs Experimental Pvt. Ltd. (Contract)* — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Python, OpenCV, C++, Django* **(Live link)**
  - Architected a native C/C++ bridge using `dart:ffi` to port OpenCV Scale-Invariant Feature Transform (SIFT) keypoint algorithms into Flutter for computer vision across Android and iOS.
  - Engineered a real-time video stream processing pipeline normalizing OS frame buffers, converting raw frames into C data structures with zero memory leaks at 30+ FPS.
  - Implemented feature matching and object recognition with statistical descriptors, transmitting match coordinates asynchronously to render real-time UI overlays.
  - Built RESTful backend APIs in Django with Python to process location-aware spatial discovery data, powering responsive cross-platform mobile interfaces for alpha launch.
- **Kia National Conference App** | *Python, SQL, REST APIs*
  - Orchestrated real-time data ingestion workflows for an event application serving 500+ attendees at the Kia National Conference in Bangkok, automating registration and venue routing.

## Technical Skills

**Data Science & Analytics:** Python, SQL, R, Statistical Modeling, Feature Engineering, NumPy, Pandas, Scikit-learn, Seaborn
**Data Engineering & Cloud:** ETL/ELT, Data Pipelines, PySpark, PostgreSQL, REST APIs, Docker, AWS, Git, Linux, Bash
**AI & Generative AI:** LLMs, Agentic AI, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Prompt Engineering, n8n
**Computer Vision & Frameworks:** OpenCV (SIFT), C++, C, Django, NextJS, ExpressJS, TypeScript, Flutter

## Achievements

**Erasmus Mundus Joint Master MATHS-DISC Fellowship** — Europe
*Issued by European Union and affiliated universities* — *Mar. 2026*

**Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships** — India
*Indian Statistical Institute and Chennai Mathematical Institute respectively* — *Jun. 2026*

## Publications

**Computable Analysis: Mathematical Foundations and Computability of Real Functions**
*Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)*